

220 kV - Core 1

General parameters

Cable length	90,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance Rct	14,00	Ohm
CT rated primary current Ipr	2.500,00	A
CT rated secondary current Isr	1,00	A
IED input burden SR	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output Sn	30,00	VA

Intermediate calculations

Total secondary burden	14,9777	Ohm
Available CT Eal	880,0000	V

REF650 Over Current Calculation

Inputs

The primary operate value	1.000,00	A
Maximum close-in fault current	31.500,00	A

Results

Criterion 1: Ealreq equation 336	352,0000	V
Criterion 2: Ealreq equation 337	554,4000	V
Required Eal	554,4000	V
Pass / fail	pass	

REF630 Over Current Calculation

Results

Actual burden of the current transformer	0,9777	VA
Approximate value of the accuracy limit factor	58,7539	
Required ALF	8,0000	
Pass / fail	pass	

220 kV - Core 1		
General parameters		
Cable length	90,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	800,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	30,00	VA
Intermediate calculations		
Total secondary burden	7,9777	Ohm
Available CT E _{al}	740,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	220,00	MVA
Nominal system voltage	220,00	kV
Transformer impedance	23,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	577,3503	A
Max. current through two main CTs and transformer I _{tf}	2.510,2186	A
Results		
E _{alreq} criterion 1: transformer rated current	172,7227	V
E _{alreq} criterion 2: through-fault current	50,0645	V
Required E _{al}	172,7227	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	50.000	A
Maximum external through-fault current I _{tmax}	50.000	A
Results		
E _{alreq} equation 601: internal close-in fault	498,6074	V
E _{alreq} equation 602: external through-fault	997,2148	V
Required E _{al}	997,2148	V
Pass / fail	fail	

150 kV - Core 1		
General parameters		
Cable length	65,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	7,7117	Ohm
Available CT E _{al}	340,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	310,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	18,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.193,1906	A
Max. current through two main CTs and transformer I _{tf}	6.628,8364	A
Results		
E _{alreq} criterion 1: transformer rated current	220,8362	V
E _{alreq} criterion 2: through-fault current	81,7912	V
Required E _{al}	220,8362	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	6.600	A
Maximum external through-fault current I _{tmax}	50.000	A
Results		
E _{alreq} equation 601: internal close-in fault	40,7177	V
E _{alreq} equation 602: external through-fault	616,9348	V
Required E _{al}	616,9348	V
Pass / fail	fail	

150 kV - Core 2		
General parameters		
Cable length	65,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	7,7117	Ohm
Available CT E _{al}	340,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	325,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	5,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.250,9256	A
Max. current through two main CTs and transformer I _{tf}	25.018,5117	A
Results		
E _{alreq} criterion 1: transformer rated current	231,5219	V
E _{alreq} criterion 2: through-fault current	308,6958	V
Required E _{al}	308,6958	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	178,9111	V
E _{alreq} equation 602: external through-fault	493,5479	V
Required E _{al}	493,5479	V
Pass / fail	fail	

150 kV - Core 3		
General parameters		
Cable length	75,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,50	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	30,00	VA
Intermediate calculations		
Total secondary burden	8,3181	Ohm
Available CT E _{al}	750,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	310,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	18,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.193,1906	A
Max. current through two main CTs and transformer I _{tf}	6.628,8364	A
Results		
E _{alreq} criterion 1: transformer rated current	238,2018	V
E _{alreq} criterion 2: through-fault current	88,2229	V
Required E _{al}	238,2018	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	266,1792	V
E _{alreq} equation 602: external through-fault	532,3583	V
Required E _{al}	532,3583	V
Pass / fail	pass	

150 kV - Core 4		
General parameters		
Cable length	70,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,50	Ohm
CT rated primary current I _{pr}	1.600,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	40,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	8,2649	Ohm
Available CT E _{al}	700,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	200,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	15,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	769,8004	A
Max. current through two main CTs and transformer I _{tf}	5.132,0024	A
Results		
E _{alreq} criterion 1: transformer rated current	119,2934	V
E _{alreq} criterion 2: through-fault current	53,0193	V
Required E _{al}	119,2934	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	206,6223	V
E _{alreq} equation 602: external through-fault	413,2446	V
Required E _{al}	413,2446	V
Pass / fail	pass	

33 kV - Core 1		
General parameters		
Cable length	13,00	m
Cable cross-section	1,50	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	14,00	Ohm
CT rated primary current I _{pr}	2.500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	15,00	VA
Intermediate calculations		
Total secondary burden	14,3889	Ohm
Available CT E _{al}	580,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	110,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.924,5009	A
Max. current through two main CTs and transformer I _{tf}	16.037,5075	A
Results		
E _{alreq} criterion 1: transformer rated current	332,2974	V
E _{alreq} criterion 2: through-fault current	184,6097	V
Required E _{al}	332,2974	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	31.500	A
Maximum external through-fault current I _{tmax}	31.500	A
Results		
E _{alreq} equation 601: internal close-in fault	181,3001	V
E _{alreq} equation 602: external through-fault	362,6003	V
Required E _{al}	362,6003	V
Pass / fail	pass	

33 kV - Core 2		
General parameters		
Cable length	110,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	20,00	Ohm
CT rated primary current I _{pr}	3.000,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	60,00	
Rated CT output S _n	30,00	VA
Intermediate calculations		
Total secondary burden	21,1905	Ohm
Available CT E _{al}	3.000,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	175,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	3.061,7060	A
Max. current through two main CTs and transformer I _{tf}	25.514,2164	A
Results		
E _{alreq} criterion 1: transformer rated current	648,7922	V
E _{alreq} criterion 2: through-fault current	360,4401	V
Required E _{al}	648,7922	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	282,5406	V
E _{alreq} equation 602: external through-fault	565,0812	V
Required E _{al}	565,0812	V
Pass / fail	pass	

33 kV - Core 3		
General parameters		
Cable length	55,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	14,00	Ohm
CT rated primary current I _{pr}	2.500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	20,00	VA
Intermediate calculations		
Total secondary burden	14,6053	Ohm
Available CT E _{al}	680,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	155,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	2.711,7967	A
Max. current through two main CTs and transformer I _{tf}	22.598,3060	A
Results		
E _{alreq} criterion 1: transformer rated current	475,2784	V
E _{alreq} criterion 2: through-fault current	264,0435	V
Required E _{al}	475,2784	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	30.000	A
Results		
E _{alreq} equation 601: internal close-in fault	169,4212	V
E _{alreq} equation 602: external through-fault	350,5265	V
Required E _{al}	350,5265	V
Pass / fail	pass	

33 kV - Core 4		
General parameters		
Cable length	80,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	18,00	Ohm
CT rated primary current I _{pr}	3.000,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	20,00	VA
Intermediate calculations		
Total secondary burden	18,8713	Ohm
Available CT E _{al}	760,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	155,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	10,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	2.711,7967	A
Max. current through two main CTs and transformer I _{tf}	27.117,9672	A
Results		
E _{alreq} criterion 1: transformer rated current	511,7514	V
E _{alreq} criterion 2: through-fault current	341,1676	V
Required E _{al}	511,7514	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	29.000	A
Results		
E _{alreq} equation 601: internal close-in fault	182,4226	V
E _{alreq} equation 602: external through-fault	364,8452	V
Required E _{al}	364,8452	V
Pass / fail	pass	

33 kV - Core 5		
General parameters		
Cable length	40,00	m
Cable cross-section	2,50	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	2,00	Ohm
CT rated primary current I _{pr}	500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	2,7010	Ohm
Available CT E _{al}	240,0000	V

RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	30,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	8,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	524,8639	A
Max. current through two main CTs and transformer I _{tf}	6.560,7985	A
Results		
E _{alreq} criterion 1: transformer rated current	85,0608	V
E _{alreq} criterion 2: through-fault current	70,8840	V
Required E _{al}	85,0608	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	8.400	A
Maximum external through-fault current I _{tmax}	8.400	A
Results		
E _{alreq} equation 601: internal close-in fault	45,3775	V
E _{alreq} equation 602: external through-fault	90,7551	V
Required E _{al}	90,7551	V
Pass / fail	pass	

CT Calculation Results & SECP Comparison

Calculation sheet	Protection function	Excel criterion 1 (V)	Excel criterion 2 (V)	Excel proposed Eal (V)	Excel status	SECP criterion 1 (V)	SECP criterion 2 (V)	SECP proposed Eal (V)	SECP status	Excel / SECP (%)	RL SECP (Ohm)	RL Error (%)
220 - 1	Transformer differential	172,72	50,06	740,00	pass	195,69	56,72	740,00	pass	88,3 %	1,0092	5,4 %
220 - 1	Line differential	498,61	997,21	740,00	fail	501,83	1.003,65	740,00	fail	99,4 %	1,0092	5,4 %
150 - 1	Transformer differential	220,84	81,79	340,00	pass	242,77	89,91	340,00	pass	91,0 %	0,7288	5,4 %
150 - 1	Line differential	40,72	616,93	340,00	fail	40,91	619,90	340,00	fail	99,5 %	0,7288	5,4 %
150 - 2	Transformer differential	231,52	308,70	340,00	pass	254,52	339,36	340,00	pass	91,0 %	0,7288	5,4 %
150 - 2	Line differential	178,91	493,55	340,00	fail	179,77	495,92	340,00	fail	99,5 %	0,7288	5,4 %
150 - 3	Transformer differential	238,20	88,22	750,00	pass	263,51	97,60	750,00	pass	90,4 %	0,8410	5,4 %
150 - 3	Line differential	266,18	532,36	750,00	pass	267,55	535,10	750,00	pass	99,5 %	0,8410	5,4 %
150 - 4	Transformer differential	119,29	53,02	700,00	pass	131,20	58,31	700,00	pass	90,9 %	0,7849	5,4 %
150 - 4	Line differential	206,62	413,24	700,00	pass	207,62	415,25	700,00	pass	99,5 %	0,7849	5,4 %
33 - 1	Transformer differential	332,30	184,61	580,00	pass	341,45	189,69	580,00	pass	97,3 %	0,3826	3,7 %
33 - 1	Line differential	181,30	362,60	580,00	pass	181,47	362,95	580,00	pass	99,9 %	0,3826	3,7 %
33 - 2	Transformer differential	648,79	360,44	3.000,00	pass	688,48	382,49	3.000,00	pass	94,2 %	1,2334	5,4 %
33 - 2	Line differential	282,54	565,08	3.000,00	pass	283,38	566,76	3.000,00	pass	99,7 %	1,2334	5,4 %
33 - 3	Transformer differential	475,28	264,04	680,00	pass	496,37	275,76	680,00	pass	95,8 %	0,6167	5,4 %
33 - 3	Line differential	169,42	350,53	680,00	pass	169,79	351,28	680,00	pass	99,8 %	0,6167	5,4 %
33 - 4	Transformer differential	511,75	341,17	760,00	pass	537,32	358,21	760,00	pass	95,2 %	0,8970	5,4 %
33 - 4	Line differential	182,42	364,85	760,00	pass	182,86	365,73	760,00	pass	99,8 %	0,8970	5,4 %
33 - 5	Transformer differential	85,06	70,88	240,00	pass	109,02	90,85	240,00	pass	78,0 %	0,7209	5,9 %
33 - 5	Line differential	45,38	90,76	240,00	pass	46,05	92,09	240,00	pass	98,5 %	0,7209	5,9 %

Entscheidungsmatrix – Transformator-Differentialschutz

A. Datengestützte Kennzahlen

Anzahl untersuchter Trafo-Diff-Fälle	10	
Minimale Abweichung Excel zu SECP	2,7 %	
Mittlere Abweichung Excel zu SECP	8,8 %	
Maximale Abweichung Excel zu SECP	22,0 %	
Übereinstimmung Pass/Fail	100,0 %	
Richtung der Abweichung	SECP höher als Excel	konservativ im untersuchten Datensatz

B. Gewichtete Qualitätsbewertung

Kategorie	Gewichtung	Punkte (0–4)	Gewichteter Beitrag	Beispielhafte Begründung	Fehlerrichtung	Maßnahme	Status
Correctness of the calculation method	20%	3	15,0%				
Safety impact of deviations	20%	4	20,0%				
Numerical accuracy of the results	20%	2	10,0%				
Documentation and presentation quality	15%	1	3,8%				
Correctness of input processing	10%	3	7,5%				
Consistency of the suitability decision	10%	4	10,0%				
Traceability and reproducibility	5%	4	5,0%				
Bewertete Gewichtung	100%						
Normiertes Gesamtergebnis	71,3 %	Beispielurteil: für Trafo-Diff grundsätzlich anwendbar, aber nur mit dokumentierter Korrektur beziehungsweise manueller Referenzprüfung.					

Entscheidungsmatrix – Linien-Differentialschutz

A. Datengestützte Kennzahlen

Anzahl untersuchter Line-Diff-Fälle	10	
Minimale Abweichung Excel zu SECP	0,1 %	
Mittlere Abweichung Excel zu SECP	0,5 %	
Maximale Abweichung Excel zu SECP	1,5 %	
Übereinstimmung Pass/Fail	100,0 %	
Richtung der Abweichung	SECP höher als Excel	konservativ im untersuchten Datensatz

B. Gewichtete Qualitätsbewertung

Kategorie	Gewichtung	Punkte (0–4)	Gewichteter Beitrag	Beispielhafte Begründung	Fehlerrichtung	Maßnahme	Status
Correctness of the calculation method	20%	4	20,0%				
Safety impact of deviations	20%	4	20,0%				
Numerical accuracy of the results	20%	3	15,0%				
Documentation and presentation quality	15%	1	3,8%				
Correctness of input processing	10%	3	7,5%				
Consistency of the suitability decision	10%	4	10,0%				
Traceability and reproducibility	5%	4	5,0%				
Bewertete Gewichtung	100%						
Normiertes Gesamtergebnis	81,3 %	Beispielurteil: für Line-Diff im untersuchten Datensatz grundsätzlich anwendbar. Die geringe konservative Abweichung ist zu dokumentieren; Grenzfälle sollten mit einer					

C. Datengrundlage – Transformator-Differentialschutz

Berechnungsblatt	Excel Kriterium 1 [V]	Excel Kriterium 2 [V]	SECP Kriterium 1 [V]	SECP Kriterium 2 [V]	Abweichung maßgebend des Kriterium	Excel Status	SECP Status
220 - 1	172,7227	50,0645	195,6871	56,7209	11,7%	pass	pass
150 - 1	220,8362	81,7912	242,7694	89,9146	9,0%	pass	pass
150 - 2	231,5219	308,6958	254,5163	339,3551	9,0%	pass	pass
150 - 3	238,2018	88,2229	263,5138	97,5977	9,6%	pass	pass
150 - 4	119,2934	53,0193	131,2000	58,3111	9,1%	pass	pass
33 - 1	332,2974	184,6097	341,4496	189,6942	2,7%	pass	pass
33 - 2	648,7922	360,4401	688,4797	382,4887	5,8%	pass	pass
33 - 3	475,2784	264,0435	496,3694	275,7608	4,2%	pass	pass
33 - 4	511,7514	341,1676	537,3154	358,2103	4,8%	pass	pass
33 - 5	85,0608	70,8840	109,0184	90,8487	22,0%	pass	pass

D. Datengrundlage – Linien-Differentialschutz

Berechnungsblatt	Excel Kriterium 1 [V]	Excel Kriterium 2 [V]	SECP Kriterium 1 [V]	SECP Kriterium 2 [V]	Abweichung maßgebend des Kriterium	Excel Status	SECP Status
220 - 1	498,6074	997,2148	501,8250	1003,6500	0,64%	fail	fail
150 - 1	40,7177	616,9348	40,9137	619,9040	0,48%	fail	fail
150 - 2	178,9111	493,5479	179,7722	495,9232	0,48%	fail	fail
150 - 3	266,1792	532,3583	267,5520	535,1040	0,51%	pass	pass
150 - 4	206,6223	413,2446	207,6225	415,2450	0,48%	pass	pass
33 - 1	181,3001	362,6003	181,4728	362,9455	0,10%	pass	pass
33 - 2	282,5406	565,0812	283,3787	566,7573	0,30%	pass	pass
33 - 3	169,4212	350,5265	169,7857	351,2808	0,21%	pass	pass
33 - 4	182,4226	364,8452	182,8643	365,7287	0,24%	pass	pass
33 - 5	45,3775	90,7551	46,0471	92,0942	1,45%	pass	pass